

Technological Revolution & World Economy

Objectives

- What is a technological revolution?
- How technology changes jobs and businesses
- How Nepal is also affected by global technology changes”

What is a Technological Revolution?

- Definition: **A period of rapid technological change that transforms industries, societies, and economies.**
- Key characteristics:
 - Introduction of new technologies
 - Rapid adoption
 - Major social, economic, and industrial impact
- When mobile phones came → people stopped using landlines
- When internet came → people stopped going to cyber cafés
-  Key idea:
- “New technology comes → old way disappears → new life starts”

Historical Technological Revolutions

- **First Industrial Revolution (1760–1840)**
 - Steam engines, mechanized production
 - Impact: Factories, urbanization
- **Second Industrial Revolution (1870–1914)**
 - Electricity, assembly lines, telegraph
 - Impact: Mass production, faster communication
- **Third Industrial Revolution (1960s–2000s)**
 - Computers, internet, automation
 - Impact: Knowledge economy, IT services

Fourth Industrial Revolution (4IR)

- Current phase: **Integration of digital, physical, and biological systems**
- Key technologies:
 - Artificial Intelligence (AI)
 - Internet of Things (IoT)
 - Robotics & automation
 - Blockchain & digital currencies
 - 3D printing & nanotechnology



Note: 4IR = “Digital + AI + IoT + Automation”

4IR – Characteristics

- **Speed:** Technology adoption is faster than previous revolutions
- **Scope:** Crosses industries, societies, and countries
- **Impact:** Affects business models, labor markets, and social structures
- **Integration:** Physical, digital, and biological systems

Global Impact of Technological Revolutions

- Increased productivity & efficiency
- Creation of new industries and jobs
- Disruption of traditional industries
- Global supply chain integration
- Innovation-driven economic growth
- **Examples:**
- Uber & Airbnb disrupting transport & hospitality
- Amazon disrupting retail globally

Technology & Digital Transformation

- Technological revolutions **enable digital transformation**
- Digital transformation = applying **4IR technologies** to innovate
- Examples:
 - AI in banking (chatbots, fraud detection)
 - IoT in manufacturing (smart factories)
 - Blockchain in supply chain (tracking products globally)

Digital Economy & Global Growth

- **Digital economy:** Economic activity based on digital technologies
- **Impact on global GDP:**
 - Faster growth in IT-enabled sectors
 - Platform economy (Uber, Airbnb, Alibaba)
- **Digital trade:** Cross-border digital services, e-commerce

Impacts on Labor Market

- Automation replaces repetitive tasks → skill shift
- Demand for **digital skills** rises
- Gig economy and freelancing grow globally
- **Nepal Example:** Freelancing platforms like Upwork and Fiverr provide jobs for youth

Impacts on Labor Market

- Some jobs disappear:
- Typist
- Telephone operator
- Some jobs appear:
- App developer
- Digital marketer
- Freelancer

Startups & Innovation

- Technology revolution encourages **start-ups**
- Rapid prototyping and global reach
- **Nepal Example:**
- eSewa, Pathao, Daraz Nepal, HamroPathshala
- **Impact:** Innovation ecosystem → economic growth & employment

Platform Economy

- Platforms leverage **network effects**
- Examples:
- Uber, Amazon, Airbnb
- Digital platforms reduce transaction costs, connect global users
- Nepal Example: Pathao connects drivers to passengers, Daraz connects sellers to buyers

Sector-wise Impact

Sector	Technology Impact	Nepal Example
Education	Online learning, LMS	HamroPathshala
Banking	Mobile banking, QR payments	eSewa, Khalti
Transportation	Ride-hailing, tracking	Pathao
Healthcare	Telemedicine	mHealth apps

Globalization & Technology

- Technology accelerates **global integration**
- Digital platforms → cross-border services
- Remote work, online education, e-commerce
- Nepal Example: Nepali freelancers working for clients abroad

Challenges of Technological Revolution

- Job displacement due to automation
- Digital divide (urban vs rural)
- Data privacy and cybersecurity risks
- Regulatory & policy gaps
- 💡 Discussion: *How can Nepal address digital divide challenges?*

Nepalese Economy & Technological Revolution

- Rapid adoption of mobile phones, internet, and digital payments
- Start-up ecosystem supported by private & government initiatives
- Digital platforms enabling youth employment and financial inclusion
- Example: Nagarik App for e-governance, Pathao for gig jobs

Future Trends

- AI-driven decision-making in businesses
- Blockchain for secure transactions and supply chains
- Smart cities and IoT-enabled infrastructure
- Nepal: gradual adoption in banking, health, and transport

Key Takeaways

- Technological revolutions reshape **industries, economies, and societies**
- 4IR technologies are core to **digital transformation**
- Nepal is gradually participating through **startups, digital payments, and e-governance**
- Digital economy growth depends on **skills, innovation, and infrastructure**